

Legacy Ledger

Legacy Ledger is a Zero-Trust, AI-powered personal finance and investment dashboard. It combines passwordless hardware biometrics (WebAuthn), functional Natural Language Processing (NLP) via Gemini, and a custom multi-currency market engine to track cash flow and synthesize portfolio recommendations.



Features

- **Zero-Trust Architecture:** Frontend intercepted by WebAuthn hardware biometrics and OTP recovery.
- **Agentic AI Parsing:** Unstructured natural language inputs are algebraically parsed into strict JSON ledger entries using Google Gemini.
- **Robo-Advisor (Lilith):** Evaluates risk appetite and financial horizons to generate tailored asset allocations.
- **Live Market Engine:** Asynchronously pulls Yahoo Finance market data and converts global commodity pricing to localized values.

Prerequisites

- Node.js (v18.0.0 or higher)
- PostgreSQL Database (Supabase recommended)
- A Google Gemini API Key

Setup & Installation

1. Clone the repository

```
git clone [https://github.com/Namith2006/legacy-ledger.git](https://github.com/  
cd legacy-ledger
```

2. Environment Setup

Create a `.env` file in the `backend` directory using the provided template:

```
cp .env.example .env
```

Fill in your `DATABASE_URL` , `GEMINI_API_KEY` , and other secrets in the `.env` file.

3. Install Dependencies

Install dependencies for both the frontend and backend:

```
# Install backend dependencies  
cd backend  
npm install
```

```
# Install frontend dependencies
cd ../frontend
npm install
```

Running the Application

Start the Backend Server

```
cd backend
npm run dev # Runs on http://localhost:5000
```

Start the Frontend Client

Open a new terminal window:

```
cd frontend
npm run dev # Runs on http://localhost:5173
```

Testing

To run the automated backend smoke test (verifies server health and database connectivity):

```
cd backend
node smoke-test.js
```

Security Notes

- Ensure `.env` is listed in your `.gitignore` file.
- In production, verify that your database connection utilizes `sslmode=require`.